Design, Simulation and Analysis of FinFET and Nanosheet FET

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1 Introduction

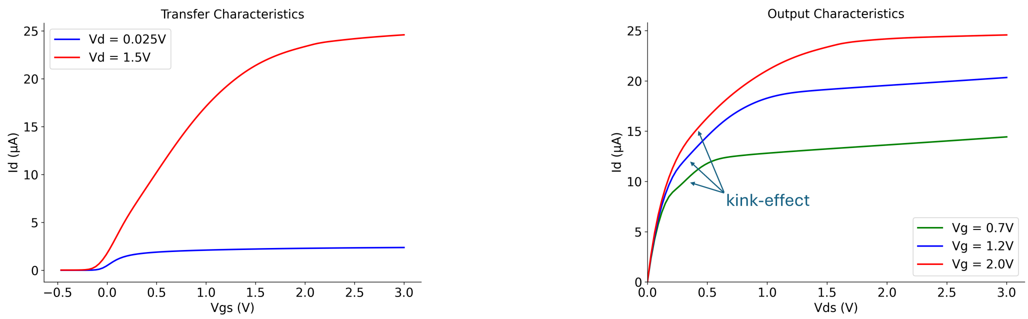
Modern transistor technologies such as the FinFETs and Nanosheet FETs do not have exact analytical expressions for their current-voltage relationships. Therefore, device characteristics and performance parameters must be evaluated using TCAD.

This report presents the design, simulation results and critical analysis of FinFETs and Nanosheet FETs. A baseline design is modified to improve its performance and better represent practical devices. The first modification is rounding the top of the channel to reduce electric field concentration in corners, improving subthreshold characteristics. The second improvement involves making a gate-all-around structure to enhance electrostatic control over the channel. The following sections describe the design of the simulated structures, followed by results and future scope.

2 Baseline design and simulation setup

The baseline design is simulated for obtaining two transfer and three output characteristic plots. The two transfer characteristics correspond to two modes of operation- linear and saturation. The output characteristics are for three different values of gate voltage.

The hydrodynamic model is used for solving the Poisson equation in the simulations. Five different SDevice command files are created to generate five characteristic curves. All the design and simulation files are shared via GitHub. The resultant plots are shown in Figure 1.



(a) Baseline Transfer Characteristics

(b) Baseline Output Characteristics

Figure 1: Plots for the baseline design

The baseline design clearly represents a depletion mode device because of negative threshold voltage. The flattening of the “knee” of the output characteristics shows the kink-effect due to corners in the fin [1]. Performance parameter extraction is performed on the resultant data using Python. Methods given in the performance parameter extraction presentation were used, along with constant current method for measuring V_{th}. The results are tabulated in Table 1.

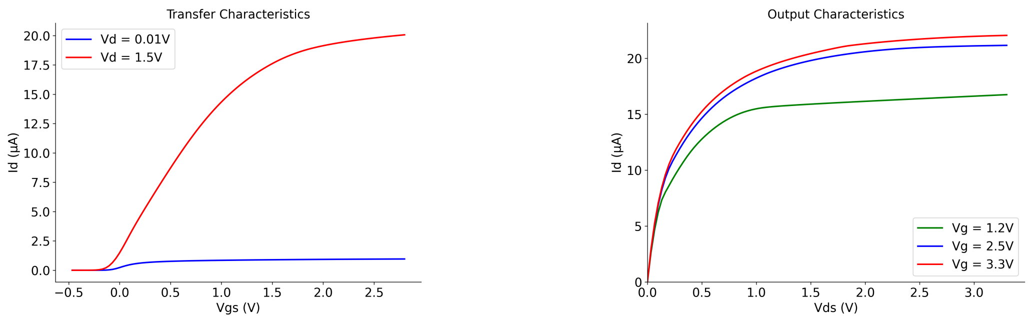
3 Rounding the top edges of FinFET

Rounding the top edges reduces the kink-effect in the transfer characteristics of a FinFET [1]. It is seen that the electron density in the corner is high for the baseline design which affects the device I_{on} and I_{off} due to leakage current [2].

The baseline version of design was modified to have rounded edges for the channel. Filleting of the edges was done with a radius of curvature of 2.5 nm to achieve rounding. Figure 2 shows the electrons density in fin of the baseline and rounded FinFETs. The electron density in the corners is high in the baseline case, which is addressed by rounding in the rounded FinFET case. The kink-effect is also reduced in the rounded case as seen in the output characteristics shown in Figure 3.

(a) Baseline (cornered fin) (b) Rounded fin

Figure 2: Electron density in the fin. A high density of electrons can be seen in the corners which is eliminated in the rounded fin. This simulation snapshot is at Vg = 3.3V and Vd = 3V.



(a) Transfer Characteristics (b) Output Characteristics

Figure 3: Plots for the Rounded FinFET with PolySilicon Gate. The output characteristics are much smoother around the "knee" of transition from linear to saturation

Further, the gate metal is changed from PolySilicon to TiN to reduce the interface state trap density [3]. While PolySilicon has a high melting temperature and facilitates tunable work function due to doping, TiN has a superior resilience to harsh etching chemicals such as HNO3, buffered HF, etc. and it can be easily deposited into fine lines as required in FinFETs [4]. The gate metal is also rounded in this case to mimic an actual process node as shown in Figure 4.

The rounded FinFET is further enhanced by giving the gate more control over the channel. This is done by reducing the oxide thickness. Reducing the thickness may increase tunneling in across it and hence a high-k dielectric, HfO2, was used in this case. Moreover, the height of the fin was increased to 20 nm to allow more current to flow through the channel.

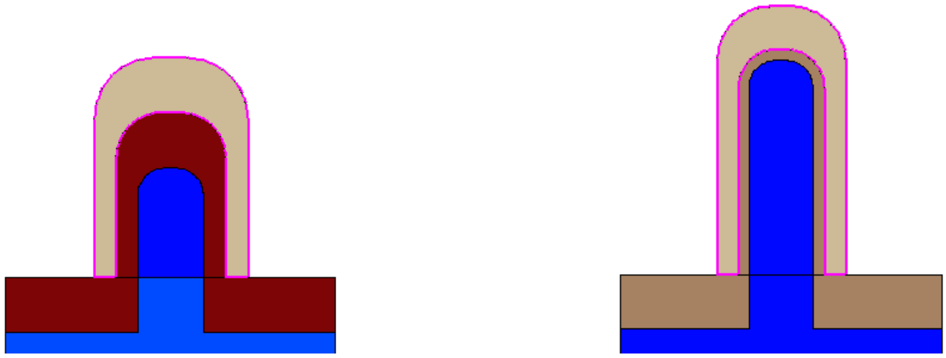
However, whether the height really contributes to the current change, needs to be verified independently, which is one of the future scopes of this project.

4 Nanosheet FET

Nanosheet FET (NSFET) is a novel method of implementing a gate-all-around (GAA) FET. It is often compared with the Nanowire FET (NWFET), which is another GAA FET technology. NSFET has a higher ON current, dimension tunability - providing enhanced I_{on} control, and higher reliability (at 7 nm node) than NWFET. The latter are advantageous in low-power applications.

TCAD simulations were performed on NSFET devices shown in Figure 5. The gate metal is TiN and the oxide is HfO2. The dimensions of the sheets are as follows $W_{NS} = 10$ nm and $T_{NS} = 5$ nm (Figure 5). And the gate length is $L_g = 8$ nm. These dimensions were chosen based on [4] and [5]. It was found in these papers that higher W_{NS} and lower T_{NS} gives higher I_{ON}, lower I_{OFF} and lower DIBL.

Since the gate wraps around the channel and oxide, and there should be physical isolation between gate and source/drain, there will be empty spaces between the gate and the electrodes. To fill this space and provide mechanical support, Silicon Nitride (Si3N4) spacers were used in the design [6].

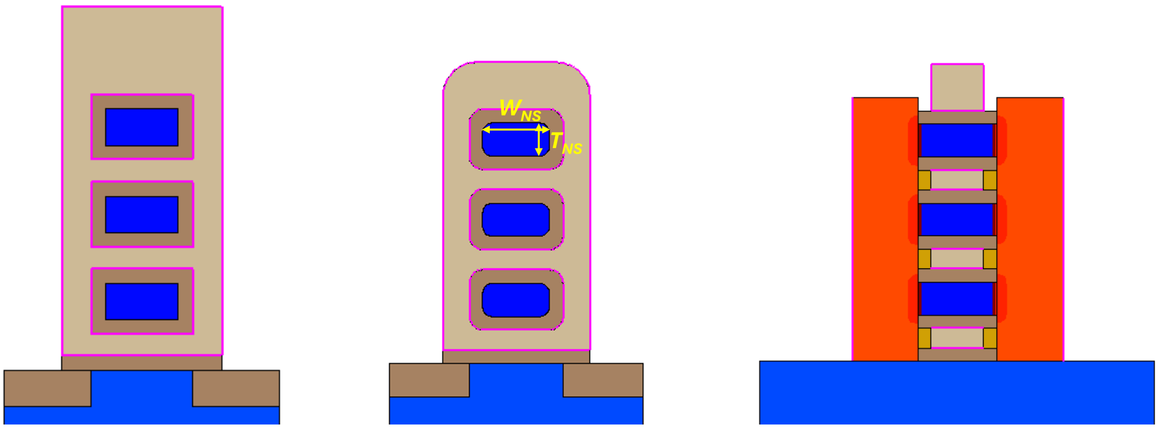


(a) TiN-rounded gate (b) Enhanced FinFET

Figure 4: Rounded FinFETs with TiN metal gate

Some designs also have an Aluminum Oxide (Al2O3) interfacial layer between Silicon channel and oxide to reduce interface trapped charges [4]. This is not included in the current design and is one of the future scopes of the project.

The edges of the NSFET were also rounded to resemble a practical device. This was done by filleting the edges of the sheets by 1.5 nm. The results of extracted parameters are tabulated and discussed in Section 5.



(a) NSFET: non-curved edges (b) NSFET: curved edges X-cut (c) NSFET: curved edges Y-cut

Figure 5: Nanosheet FET designs. Curved edges were introduced to simulate realistic models. The Y-cut for both cases looks similar and hence it is shown only for the latter. The orange-brown blocks in (c) represent Si3N4 spacers.

5 Results and discussion

The results of simulations are summarised in Table 1. A closer look at each parameter is given in the following.

5.1 Threshold Voltage

The V_{th} values for the PolySilicon gate transistors is negative, indicating that they are depletion mode devices. This is because the work-function of the heavily doped n-type PolySilicon is low. This places the flat-band voltage at a negative value, pulling down the V_{th}. For the TiN gate devices, we see that the threshold voltage is almost a half volt in most of the cases. This is a result of high workfunction of the metal. This is a tunable parameter and is set to 4.6 in the design files.

5.2 Subthreshold Swing

The baseline design has a SS of 118.6 mV/dec, nearly double the theoretical minimum of 60 mV/dec. Rounding the gate immediately reduces it to roughly 85 mV/dec, much closer to the minimum. This suggests that the sharp fin corners have a high electric field and collect electrons in the corners, reducing gate's control over the channel.

The enhanced rounded FinFET has an SS of 62.8 mV/dec very close to 60 mV/dec limit. The thinning of oxide and replacing it to a high-k dielectric substantially reduces the current in subthreshold region. The two GAA-FET configurations also have a low SS ~ 64 mV/dec, demonstrating clear advantage of wrapping the channel by gate on all sides.

5.3 DIBL

The DIBL values are interesting. DIBL slightly worsens when the channel is rounded, without changing the gate material or rounding the gate itself. This could be because rounding alone doesn't improve the effective oxide capacitance seen by the channel. It might reduce it, thereby marginally weakening gate control relative to drain influence.

Changing the gate to metal improves DIBL to 62.1 mV/V. This is due to TiN's higher compatibility with the dielectric [4]. The enhanced rounded FinFET shows the drastic effect of having high-k, thin dielectric, resulting in a DIBL of 7.1 mV/V. The NSFET designs achieve a DIBL of $17 - 20$ mV/V, and can be further improved by increasing the "parasitic channel height", which is like introducing a fin below the nanosheets [7].

5.4 Transconductance and ON current

It is desirable to have a high value of transconductance for a semiconductor device. The low $g_{m,lin}$ of the Rounded PolySilicon device, 0.101 mS/μm, despite its moderate $g_{m,sat}$ (0.612 mS/μm) suggests that in the linear regime this device is significantly affected by source/drain resistance or poor channel charge density at low drain bias. This might be linked to its depletion-mode operation producing low inversion charge at the operating gate voltage.

On the other hand, the $g_{m,sat}$ increases to 1.611 mS/μm in NSFET. This is due to better gate coupling and wider effective channel width. Increasing the width further can improve the conductance of the device. Also, note that the ratio of $g_{m,sat}$ to $g_{m,lin}$ depicts velocity saturation. Since, this value drops for the NSFET, it proves better transport efficiency in this case.

The ON current I_{ON} is maximum for the case of NSFET without rounding. This is due to wider effective channel area for current flow. The I_{ON} of rounded FinFET is lower than the baseline version, but that is compensated for a very low I_{OFF} of the rounded FinFETs. We also see an increase in I_{ON} when going from the rounded TiN FinFET to the enhanced rounded FinFET.

5.5 The ON and OFF current ratio

The I_{ON}/I_{OFF} ratio determines the working of the device. Both the PolySilicon devices (baseline and rounded) show a ratio $\sim 12 - 13$, this is because they are depletion mode devices and have a finite current at zero V_{GS}. To measure the true I_{OFF}, current at $|V_{GS}| \ll |V_{th}|$ needs to be measured. After measuring that, we get $I_{ON}/I_{OFF} = 1.1e + 03$ and $4.6e + 05$ for the baseline and rounded poly cases.

The introduction of TiN gate improves the ratio roughly ten folds. Nanosheet FETs have a superior ratio compared to their FinFET counterparts. This shows that the NSFET not only has a better on current but also lower leakage, again proving gates aided control over the channel. The best ratio is obtained for the enhanced rounded fin FinFET, suppressing the I_{OFF} to pA range. Similar enhancements in NSFET could give better results in that case too.

Table 1: Extracted device parameters. (*I_{OFF} at $V_{GS} = 0$, Sec. 5.5; g_d is for median V_{GS}, Sec. 5.6)

Model	$g_{m,\mathrm{lin}}$ **(mS/µm)**	$g_{m,\mathrm{sat}}$ **(mS/µm)**	$V_{th,\mathrm{lin}}$ **(V)**	$V_{th,\mathrm{sat}}$ **(V)**	**SS (mV/dec)**	**DIBL (mV/V)**	$g_{d,\mathrm{sat}}$ **(µS/µm)**	I_{on} **(mA/µm)**	I_{off} **(mA/µm)**	$I_{\mathrm{on}}/I_{\mathrm{off}}$
Baseline	0.2567	0.7425	−0.074	−0.174	118.639	67.722	30.82	0.822	6.169e-02*	1.332e+01*
Rounded (PolySi gate)	0.1009	0.6118	−0.041	−0.159	85.061	79.635	6.249	0.680	5.501e-02*	1.236e+01*
Rounded (TiN gate)	0.2353	0.6038	0.485	0.393	85.151	62.119	23.27	0.532	2.125e-07	2.506e+06
Enhanced Rounded	0.5233	1.4250	0.440	0.430	62.847	7.136	2.528	0.615	8.819e-10	6.968e+08
NSFET (no rounding)	0.7181	1.6110	0.458	0.431	64.996	17.761	3.905	1.002	1.493e-08	6.712e+07
NSFET (with rounding)	0.6740	1.5370	0.459	0.435	64.565	20.308	5.519	0.827	1.176e-08	7.034e+07

5.6 Output Conductance

The output conductance represents the channel length modulation effect and should be zero ideally. The baseline device has a $g_{d,sat}$ of 0.031 mS/μm and $g_{m,sat}$ of 0.743 mS/μm, resulting in a low intrinsic gain of $A_v = g_m/g_d \approx 24$. It increases to 98 for the rounded PolySilicon FinFET. The gain for the NSFET case are 413 and 278 in the two cases, these values are consistent with results in [4]. The best value from our simulations is for the enhanced rounded FinFET, which has a gain of $A_v = 475$.

Table 2 shows the value of g_d at different V_{GS} for a given model. The output characteristics were obtained at different values of V_{GS} for different models, to be close to the true V_{DD} they may vary in practical applications (It is slightly high for the rounded poly case, as these simulations were performed at an earlier stage). These voltages might heat up the device, and can be reduced in the future.

Table 2: Output conductance g_d ($\mu S/\mu m$)

Model	g_d	V_{GS} **(V)**	g_d	V_{GS} **(V)**	g_d	V_{GS} **(V)**
Baseline	31.400	0.7	30.820	1.2	11.710	2.0
Rounded (Poly Gate)	17.170	1.2	6.249	2.5	11.100	3.3
Rounded (TiN Gate)	22.310	0.7	23.270	1.2	15.370	2.0
Enhanced Rounded	3.912	0.7	2.528	1.2	1.178	2.0
NSFET	6.163	0.5	3.904	1.2	0.969	2.5
NSFET (rounded)	6.242	0.7	5.520	1.0	2.486	1.5

6 Conclusion and future scope

TCAD was used for the design and simulation of six different advanced semiconductor devices. A rudimentary baseline design was modified for two structural changes- rounding of the top edges of the fin and wrapping the gate around the channel, creating Nanosheet FETs.

It was seen that corner rounding improves FinFET parameters and removes the kink-effect but worsens the NSFET parameters. This is because rounding in a nanosheet effectively reduces the usable conducting cross-sectional area of each sheet, as the rounded edges contribute less current than the full rectangular channel. Unlike in the FinFET case, where rounding removes corner electrons, nanosheet rounding reduces a real physical contributor to I_{on}.

To summarise, changing the metal material changes the threshold value and takes the device from depletion to enhancement mode. Reducing the width of the oxide and increasing the dielectric constant massively improves the performance parameters. When the fin width is more than the half-gate length, ($W_{fin} > 0.5L_g - 6t_{ox}$), increasing the height of the fin might improve the characteristics [3]. NSFET reduces the short channel effects by providing more control to the gate.

As a future scope, the NSFET oxide can be made thinner, the width can be increased further, and "parasitic channel height" can be introduced. It would be interesting to simulate punch-through effect, device breakdown and Self-Heating Effects at higher voltages. Given more time, it would be nice to tune parameters like the metal workfunction, oxide dielectric constant, temperature and gate/channel length.

References

[1] Mirko Poljak, Vladimir Jovanović, and Tomislav Suligoj. Suppression of corner effects in wide-channel triple-gate bulk finfets. *Microelectronic Engineering*, 87(2):192–199, 2010.

[2] Suman Lata Tripathi and Roshan Mishra. Design of 20 nm finfet structure with round fin corners using side surface slope variation. *Journal of Electronic Devices*, 2013.

[3] K Fobelets. Advanced semiconductor devices summary course notes. *[unpublished]*, 2017.

[4] Sresta Valasa, Shubham Tayal, Laxman Raju Thoutam, J. Ajayan, and Sandip Bhattacharya. A critical review on performance, reliability, and fabrication challenges in nanosheet fet for future analog/digital ic. *Micro and Nanostructures*, 2022.

[5] V. Jegadheesan, K. Sivasankaran, and Aniruddha Konar. Impact of geometrical parameters and substrate on analog/rf performance of stacked nanosheet field effect transistor. *Materials Science in Semiconductor Processing*, 93:188–195, 2019.

[6] Khwang-Sun Lee and Jun-Young Park. Inner spacer engineering to improve mechanical stability in channel-release process of nanosheet fets. *Electronics*, 10:1395, 06 2021.

[7] et. al. Yunho Choi. Simulation of the effect of parasitic channel height on characteristics of stacked gate-all-around nanosheet fet. *Solid-State Electronics*, 164:107686, 2020.